Module 06 [Data Structure with STL]
(for Week 7)

Topic 1 [STL: stack class]
Problem statement: Write a menu program to create and manipulate Stack using `stack` class and perform the following operations using the specified method.

i) use `push()` method to push data
ii) use `pop()` method to pop data
iii) use `top()` method to display top element
iv) use `empty()` method to check whether stack is empty or not

```
 ** Stack **
 1. Push
 2. Pop
 3. Display()
 4. Exit
    Enter your option:
```

Topic 2 [STL: queue class]
Problem statement: Write a menu program to create and manipulate Queue using `queue` class. Perform the following operations using the specified method.
i) use `push()` method to enqueue data
ii) use `pop()` method to dequeue data
iii) use `front()` method to display front element
iv) use `back()` method to display rear element
v) use `empty()` method to check whether queue is empty or not

```
 ** Queue **
 1. Enqueue
 2. Dequeue
 3. Display Front
 4. Display Rear
 5. Exit
    Enter your option:
```

Topic 3 [STL: vector class]
Problem statement: Write a menu program to create and manipulate linked list using `vector` class and use the following methods
i) **`push_back()`**
ii) **`push_front()`**
iii) **`begin()`**
iv) **`end()`**

v) **pop_back()**
vi) **pop_front()**
vii) **insert()**
viii) **erase()**

```
 ** Linked List **
 1. Insert
 2. Delete
 3. Update
 4. Search
 5. Exit
    Enter your option:
```

Topic 3.1 [STL: vector class]: Add more feature to the menu of Topic 3.

Module 07 [UML+Java Basics]
(for Week 8: 19-23/7/2025)

Topic 1 [UML]
Problem statement: Draw the UML diagram for the following class

```
class Test{
  private: int x;
  protected: int y;
  public: int z;
  public:
    void SetData(int a,int b,int c){
      x=a;y=b;z=c;
  }
  float getAverage(){
   Return (x+y+z)/3.0;
 }
```

Topic 2 [UML: Aggregation]
Problem statement: Create two classes Doctor and Patient to store the information of doctors and patients respectively. Now write codes to create association between these two classes by aggregation.
[note: if Patient class is destroyed then Doctor class is still exist]

class Doctor{
//write your code
}
	class Patient{
	//write your code
	}
		int main(){
		//write your code
		}

Topic 3 [UML: Composition]
Problem statement: Create two classes Doctor and Patient to store the information of doctors and patients respectively. Now write codes to create association between these two classes by composition.
[note: if Doctor class is destroyed then Patient class is also destroyed]

class Doctor{
//write your code
}
	class Patient{
	//write your code
	}
		int main(){
		//write your code
		}

Topic 4: [**java Programming Basics**]
Problem Statement: Write java programs to do the following

i) Display your name and address in two lines
ii) Take two integers as inputs and prints the bigger one
iii) Input a series of 10 numbers into an array ax[] and finds the largest, smallest and average of these numbers. Also write code for searching and sorting.

Topic 5 [**java Programming private method**]
Problem Statement: Verify that private methods can be used only inside the class but it cannot be used from outside the class.

Topic 6 [**java Programming static method/member**]
Problem Statement: Verify that static methods can access static method/member but cannot access to non-static method/member.

Topic 7 [**java Programming**]
Problem Statement: Write a Java Program to initialize radius of a circle using constructor/Setter and calculate area of the circle.